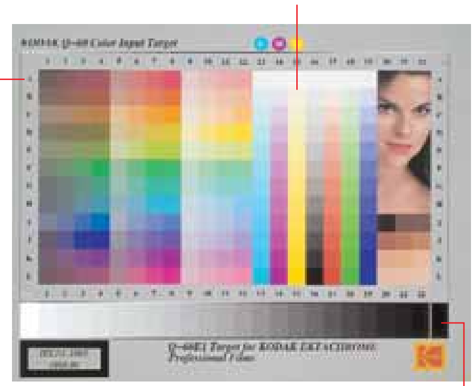
TEST PROCESS

We ran a battery of tests that evaluated a range of characteristics of the scanners besides just the image quality. We considered four factors in determining the performance of each scanner:



■ **Image Quality:** Since the prime function of a scanner is to pick up images and convert them into digital format, image quality is one of the most important parameters for evaluating flatbed scanners. Two reference images were used to test the image quality—the IT8.7 colour card, which is a standard image used worldwide for colour correction and calibration, and an A4-size photograph that helped determine the accuracy of the scanner in picking out details and subtle variations in colour, hue, brightness, and resolution.

■ **Resolution:** To determine the maximum resolution at which each scanner could pick out the finer details, we used a resolution target that consisted of five squares forming bar patterns, created at different dpi ratings.

■ **Speed:** We logged the time taken to scan a range of different images, and at different resolutions and colour depths. Additionally, the time taken to scan text documents and photographs in full colour and gray scale modes was also logged.

■ **Features:** We also considered build quality, ability to scan transparencies, documentation, buttons for one-touch scanning operations, and provision for upgrading to an Automatic Document Feeder.

The preview speed depends a lot on the technology used, the implementation of the optics used in the scanner, and the software that controls these scanning elements.

Next, we scanned the document in black-and-white (line-art) mode and then in gray scale mode at 200 dpi. The document was scanned directly to Photoshop from the Twain drivers of the respective scanners.

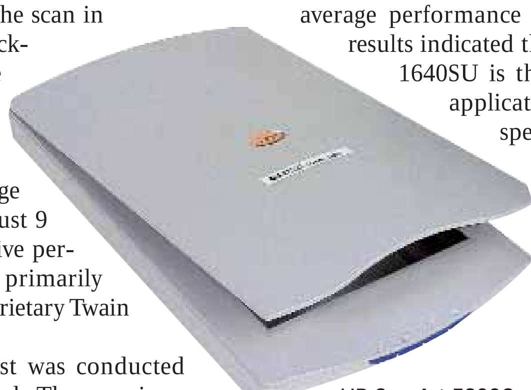
The HP ScanJet 5300C was the best among the pack in this test—it took just 8.31 seconds to scan the document in black-and-white mode, and 9.25 seconds for the scan in gray scale mode. In black-and-white mode, the HP ScanJet 5300C scanned the image, opened Photoshop and displayed the image in the application in just 9 seconds. This impressive performance could be primarily attributed to HP's proprietary Twain drivers.

The final speed test was conducted using the test photograph. The scanning was done at 300 dpi in full colour mode and subsequently in gray scale mode, using the maximum bit-depth offered by the scanner.

In gray scale mode, the HP ScanJet 5300C emerged the winner with a timing of 13.5 seconds. The Umax Astra 2400S, which is a SCSI scanner, came in second by clocking 15 seconds.

The fastest scanner in full colour mode was the Epson Perfection 1640SU. It scanned the test photograph in just 25.31 seconds. The first runner-up was the Umax Astra 3400 with a timing of 31 seconds. This scanner, however, gave an average performance in the other tests. The results indicated that the Epson Perfection 1640SU is the best suited for office applications because of its speed in scanning the photograph, although it was not quite as fast in scanning the text document.

The Umax Astra 3400 is the perfect buy if you are looking for an inexpensive scanner for home use.



HP ScanJet 5300C

Contents